

SCHOOL OF COMPUTING AND INFORMATICS

Course Code	CCE2233	Course Name	Requirements Engineering
Lecturer	Mdm. Nadiah Arsat		
Semester	2, 2025/2026		
Lab Session	Module 7		
Course Learning Outcome	CLO 2 - Describe software requirement engineering processes using appropriate tools and techniques (C2, PLO6).		
Student Name	Chit Ko Ko		
Student ID	AIU24102180		
Lab Group	B		

PART A: Short Answer Questions

1. What is system context?

System context is the environment surrounding a system, including all external entities that interact with it.

2. List TWO aspects that influence system context.

- Stakeholders (users, organizations)
- External systems and regulations

3. State the purpose of defining system and context boundaries.

To identify what is inside the system and what is outside the system.

4. Define system boundary.

The system boundary defines the functions and components that belong to the system.

5. Define context boundary.

The context boundary separates the system and its environment from unrelated external entities.

6. What is the consequence of incorrect system context?

It may lead to incorrect requirements and system failures.

7. What are sources and sinks?

Sources provide data to the system, while sinks receive data from the system.

8. What is the grey zone between system and system context?

The grey zone refers to components whose ownership or responsibility is unclear between the system and its environment.

9. Name TWO techniques to document system context.

- Context Diagrams
- Use Case Diagrams

10. What is tool support in Requirements Engineering?

Tool support refers to software tools used to assist requirements engineering activities.

11. Give TWO examples of general tool support.

- Microsoft Word
- Microsoft Excel

12. State ONE advantage of using general development tools for requirements management.

They are easy to use and widely available.

13. What are modelling tools used for?

Modelling tools are used to create visual representations of systems and requirements.

14. State TWO properties of modelling tools.

- Graphical representation of requirements

- Support for analysis and communication

15. List TWO categories of requirements management tools.

- General-purpose tools
- Specialized requirements management tools

PART B: Case Study Questions

Case Study: Student Management System

1. Identify THREE elements of the system context.

- Students
- Lecturers
- Payment System

2. Differentiate between system and system context.

The **system** is the Student Management System itself that manages student information and services. The **system context** consists of external entities such as students, lecturers, payment systems, and university regulations that interact with the system.

3. Give ONE example of system boundary.

Student registration and course enrollment functions are inside the system boundary.

4. Give ONE example of context boundary.

The external payment processing service is outside the system boundary but within the system context.

5. Explain the grey zone in this case.

Payment processing may be partially controlled by the university and partially by an external provider, making responsibility unclear.

6. Identify ONE source and ONE sink.

- Source: Student submitting registration information.
- Sink: Lecturer receiving student course information.

7. Suggest ONE suitable general tool and justify.

Microsoft Word can be used because it allows requirements to be documented, edited, and shared easily.

8. Explain how modelling tools can be used in this system.

Modelling tools can create use case diagrams, context diagrams, and process models to visualize system requirements and interactions.

9. Why is traceability important?

Traceability helps track requirements from their origin through development and testing, ensuring all requirements are implemented correctly.

10. Evaluate a tool using ONE evaluation view.

Usability View: Microsoft Word is easy to learn and use, making it suitable for documenting and managing requirements efficiently.